A Dual-UNet With Multistage Details Injection for Hyperspectral Image Fusion

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***Abstract*— Enhancement of hyperspectral image (HSI) resolution is significant for better application in practice. In this article, a dual U-Net (D-UNet) is proposed to improve the spatial resolution of HSI. The whole network contains two parts. One is the detail extraction network, whose network architecture is encoder–decoder and mainly extracts various spatial features from multispectral images (MSIs). Another is the spatio-spectral fusion network (SSFN), which aims at injecting the features from the detail extraction network into HSI for better reconstruction. Furthermore, in the primary stage of the whole network, a novel multiscale spatio-spectral attention module (MSSAM) is utilized to pay more attention to important features at different scales. Considering the complex ground scenes, the features of different scale and depth are continually extracted and fused in the whole network. The experimental results show that the proposed method is more effective compared with the state-of-the-art methods.**

***Index Terms*— Attention module, hyperspectral image (HSI), multiscale, U-Net.**

I. Introduction

HYPERSPECTRAL images (HSIs) covering wide-range wavelength own dozens to hundreds of bands with small wavelength interval, which can discriminate many similar materials and is suitable for many remote-sensing applications [1], such as classification [2], [3], [44], object recognition [4], detection [5], disasters [6], and biodiversity [7]. However, in comparison with high spatial resolution of multispectral images (MSIs) or panchromatic images, low spatial resolution of HSIs limits the understanding about ground materials, such as high spatial resolution ecological monitoring, fine classification of surface objects, or mapping of urban land.

Therefore, to enhance HSIs spatial resolution by fusing the high-frequency information of MSIs aroused great enthusiasm. In early years, pan-sharpening methods are directly introduced to fuse HSI and MSI, such as component substitution (CS) [8], multiresolution analysis (MRA) [9], and model-based methods [10]. These categories, however, hardly consider the spectral characteristics of HSI, resulting in spectral distortion in the fused image. Then, a set of HSI and MSI fusion methods are proposed to get better performance, including variational model-based framework [11]–[13], [28], matrix factorization [14]–[16], [27], and tensor factorization [31]–[33]. In variational model-based framework, image fusion is regarded as an ill-posed problem, thus an alternating direction method of multipliers is often used to estimate the unknown spatial and spectral degradation functions for solution of non-convex problem [13]. Some handcraft priors, of the image, for example, sparse and low-rank priors, are designed to better restrain the solution of fusion. Considering the correlation between HS and MS images, non-negative matrix factorization [14] utilizes the endmember matrix of HSI and abundance map of MSI to reconstruct the latent high-resolution hyperspectral images (HR HSIs). Besides, sparse representation is also applied to achieve matrix factorization, with the assumption that each pixel comprises of a small amount of pure endmembers compared to an abundant endmember matrix [15]. Compared with matrix factorization, tensor factorization considers the image as a tensor without destroying the spatio-spectral structure. It may be better to solve reconstruction problem in terms of tensor. A classic method, called Tucker decomposition, assumes that HR HSI can be estimated by the product of a core tensor and the factor matrix along three modes [31]. Thus, coupled sparse tensor factorization (CSTF) can estimates three high-resolution (HR) dictionaries and core tensor by constructing the spatial and spectral degradation between HR HSI and two input observation images based on tensor factorization [33]. Spatial–spectral reconstruction network (SSR-Net), another three-layer network with skip connection, achieves great performance by gradient loss on features of different layers [21]. Different from 2-D convolution, 3-D convolution can concentrate on the correlation of bands so that the spectral information of HSI can better be preserved [22]. Recently, deep-learning methods jointed with variational model-based methods are proposed to further restore spatial and spectral details [23], [24]. The solution process of the variational model guides the construction of the network and makes deep-learning-based methods interpretable. For example, the unfolded Proximal gradient algorithm is formulated by convolution neural network (CNN) to solve the fusion model of HSI and MSI [24].

Although the deep-learning-based methods achieve excellent performance, there is still improvement room in spatial details. Most deep-learning methods regard the concatenation of HSI and MSI along spectral channel as input of the network,

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which does not take the latent multiscale spatial information into full consideration. Multiscale MSI down-sampled by Bicubic was injected into each stage of decoder for spatial information enhancement [19], but the down-sampling operator may lose many important high-frequency information. The only details injection in the last few layers of the network may result in insufficient compensation of spatial information. Two-branch architecture [25] is proposed to, respectively, extract and gradually merges features from MSI and HSI but neglected the basic and shallow features of MSI.

Motivated by the above-mentioned problems, the article designed a novel dual U-Net, one of which outputs the hierarchical spatial features from MSI by each stage of encoder–decoder and another receives the spatial features from each stage of the detail extraction network and gradually integrates these multiscale features for reconstruction. The whole fusion process contains three steps: direct fusion, slow fusion, and late fusion. As for direct fusion, MSI and HSI are concatenated along spectral channel and input into a specially designed multiscale spatio-spectral attention module (MSSAM) in the front of the spatio-spectral fusion network (SSFN) for strengthening more valuable information. At once, this step can fuse a number of basic and important features. Then the slow fusion is followed, which further extracts and fuses spatial and spectral features at different scales. During this step, the hierarchical features output from each stage of the detail extraction network are concatenated with the output of corresponding stage in SSFN along channel for subsequent fusion. Notably, the hierarchical features output from each stage of encoder in both networks are concatenated with input of corresponding stage in decoder of SSFN by a skip connection for context information aggregation. Except that, the hierarchical features are also input into the next stage of the detail extraction network. As for late fusion, an up-sampled low-resolution (LR) HSI, and the outputs from U-Net and detail extraction network are added at pixel level for final reconstruction. To better maintain spectrum, L1 loss joint with spectral gradient loss is applied to guide the network to restore images. The main contributions in this article are as follows, in general.

1) A novel dual U-Net network is proposed to fuse MSI and HSI. The detail extraction network concentrates on rich spatial features with different scales and depths from MSI. SSFN injects these spatial features into HSI stage by stage for fine-grained reconstruction.
2) MSSAM is designed for multiscale information mining and fusion. This module combines inception to acquire multiscale features and utilizes the attention coefficients calculated on different scales to guide the network focus on various types of ground objects.
3) According to the visual and quantitative results on simulation and real experiments, the proposed network is more effective than the state-of-the-art methods. For instance, the specially designed network architecture can restore more image details. Meanwhile, spectral information is well maintained by spectral gradient loss.

The rest of this article is organized as follows. Section II introduces the proposed method and its interior structure at length. The experimental results and analysis are represented in Section III, followed by a conclusion and discussion about future work in Section IV.

II. Methodology

In this section, we elaborate on the proposed approach and its components. The overall flowchart of the method is shown in Fig. 1. LR HSI denotes by $H_l \in R^{hw\times B}$ (h, w, and B represent the height, width, and band number, respectively), MSI denotes by $M \in R^{HW\times b}$ and HR HSI denotes by $H_R \in R^{HW\times B}$. $H = h \times r$, $W = w \times r$, r is the image size ratio between MSI and HSI. MSSAM is devoted to help the network pay attention to multiscale information. The detail extraction network purely extracts abundant spatial information from MSI in different depths and scales by the encoder–decoder. SSFN is responsible to integrate spatial information from each stage of the detail extraction network into HSI for enhancement of details. To further restore the high-frequency information, the final outputs of the detail extraction network and SSFN are added to LR HSI at pixel level for the reconstruction image, which is actually a residual connection as shown in Fig. 1.

A. Multiscale Spatio-Spectral Attention Module

Inception architecture [26] is good at obtaining multiscale information, so a modified inception is introduced into the attention module. The architecture of MSSAM is shown in Fig. 2. First, one convolution layer extracts shallow features, and a set of convolution layers with different receipt fields, that is, 3×3, 5×5, and 7×7, are used to acquire features at three scales. Leaky ReLU takes place of ReLU as the activation function, following the convolutions. In order to more fully mine the multiscale features of the original images, the previous features at three scales, that is, $\mathrm{Fea}_3, \mathrm{Fea}_5, \mathrm{Fea}_7$ are endowed with weight factors and fused in pairs to get three new features. The channel number of fusion features remains the same as before. γ_j is the weight factor which can be learned and pay relatively more attention to the important information. It cannot fully consider complex remote-sensing scenes only relying on the direct fusion of multi-scale features fusion. Thus, it is necessary to enhance the extracted multi-scale features. Through pairwise fusion, the features at different scales are merged so that more combinations of features with different scales can be obtained.

After that, three new features are fused for the multiscale synthetic feature block, that is, Fea, which can fully extract scale features in the two-step fusion. The formula is given by

$$F_s = \sigma\left(W_s * \left(\mathrm{cat}\left(M, H_{\mathrm{up}}\right)\right)\right) \tag{1}$$

$$\mathrm{Fea}_3, \mathrm{Fea}_5, \mathrm{Fea}_7 = f_m(F_s) \tag{2}$$

$$\mathrm{Fea} = f_1(\mathrm{Fea}_3, \mathrm{Fea}_5, \mathrm{Fea}_7) \tag{3}$$

where $*$ is convolution operator, cat is concatenation operator. f_m are a set of convolution layers to obtain multiscale features, that is, $\mathrm{Fea}_3, \mathrm{Fea}_5, \mathrm{Fea}_7$. f_1 is fusion operator.

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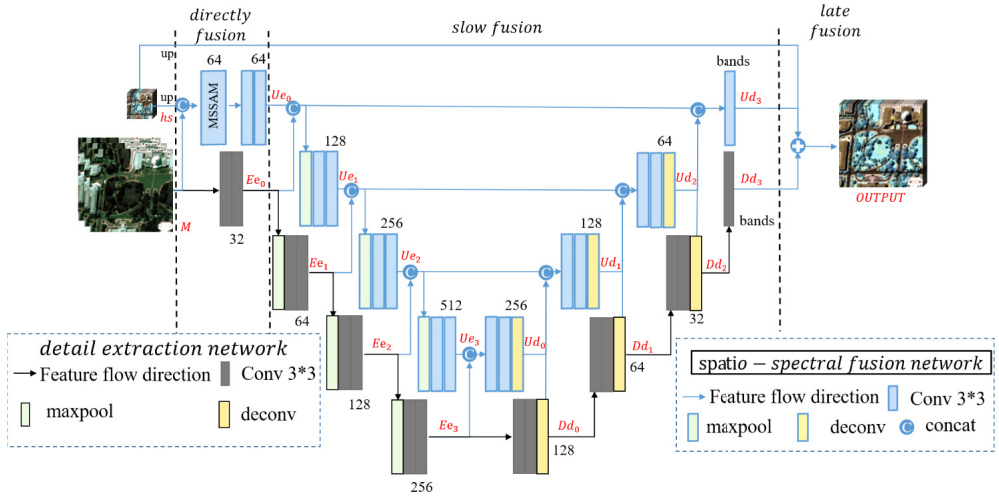


Fig. 1. Flowchart of the proposed network.

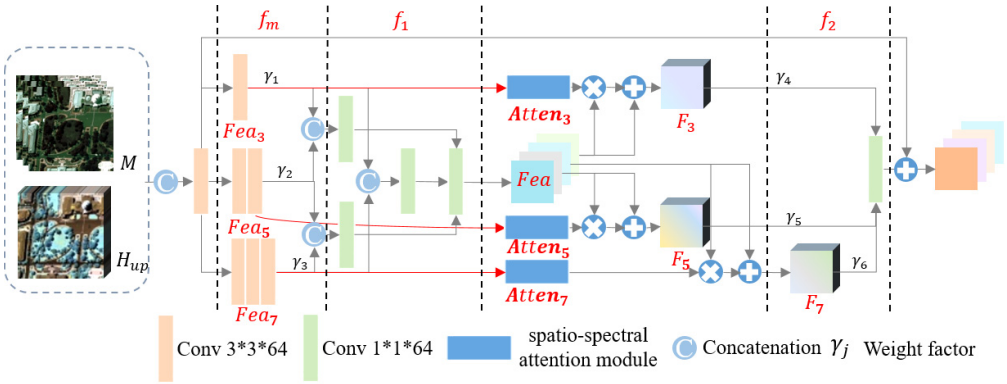


Fig. 2. Flowchart of MSSAM. Fea_i is the different-scale features, $i \in \{3, 5, 7\}$. f is the convolution function. Atten_i is the attention coefficients. F_i is the enhanced features. γ_j is the weight factor, $j \in \{1, 2, 3, 4, 5, 6\}$.

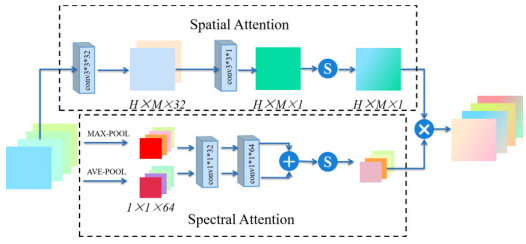


Fig. 3. Architecture of the spatio-spectral module.

Different from previous spatio-spectral attention module [39], MSSAM utilizes three attention coefficients calculated on different-scale features, that is, $\text{Fea}_3, \text{Fea}_5$, and Fea_7, to acquire three synthetical feature blocks focusing on information at different scales. The architecture of this spatio-spectral attention module is displayed in Fig. 3. In the spatial attention module, the features are input into first convolution layer to compress channels in half. After that, the second convolution layer further compresses the channel information to change the channel to 1. One sigmoid activation function outputs the spatial attention coefficient. In spectral attention module, a max-pooling and an average-pooling, respectively, compress spatial information to get two vectors with different focuses. Subsequently, the first convolution layer continues to compress the channel information of two vectors in half. Then, the second convolution layer is employed to restore the number of channels to the target number. To integrate the information contained in vectors, two vectors are added at pixel, followed by one sigmoid activation function for spectral attention module. Finally, the spatio-spectral attention coefficient is acquired by multiplying the spatial attention coefficient with spectral attention coefficient. Three coefficients at different scales do not act on input features but the synthetic feature block obtained later, that is, Fea in Fig. 2. Finally, the enhanced synthetic feature blocks are weighted and fused by one convolution layer, followed by a skip connection to add them to shallow features. The formula is as follows:

$$\text{att}_s = \text{Atten}_s(\text{Fea}_s) \tag{4}$$

$$F_s = \text{att}_s * (\text{Fea}) + \text{Fea} \tag{5}$$

$$F = f_2(F_3, F_5, F_7) \tag{6}$$

where F is the output of the whole attention module, F_s is the shallow features, σ is Leaky ReLU, and Atten_s is the

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spatio-spectral attention module whose output is multiscale attention coefficients. $s \in \{3, 5, 7\}$. F_s is the synthetic features enhanced by Atten_s. f_2 is the fusion operator. Through the MSSAM, the multiscale information can be mined to a great extent. Meanwhile, it can effectively help the network learn and reconstruct more spatio-spectral details. The related experiment is shown in the ablation study.

B. Detail Extraction Network

This part introduces the detail extraction network. It is actually a spatial information compensator for the SSFN. Generally, there are amounts of ground objects with various shapes and sizes in remote-sensing images. In consequence, it is significant to strengthen the property of the network suitable for different remote-sensing scenes. Encoder–decoder provides a good strategy to dig out multiscale information. Encoder exploits max-pooling to expand receptive filed and realize layer representation. The architecture is shown in Fig. 1. MSI is input into two convolution layers with a size of 3×3 whose number of output channels is 32 for initial features. Then, each of three down-sampling layers owns a max-pooling with a stride size 2, followed by two convolutions. In order to keep the information as far as possible from being lost by the pooling layer, the number of convolution kernels is twice as the one in previous stage. The numbers in Fig. 1 represent the number of convolution kernels per layer. The formula of encoder denotes by

$$Ee_n = \begin{cases} \sigma\left(W_n^2 * \sigma\left(W_n^1 * M\right)\right), & n = 0 \\ \sigma\left(W_n^2 * \sigma\left(W_n^1 * \text{maxpool}(Ee_{n-1})\right)\right), & n \in \{1, 2, 3\} \end{cases} \tag{7}$$

where Ee_n is the result of nth stage in the encoder, W_n^1 and W_n^2 are the first and second convolutions of the nth stage which is used to extract multiscale features. $n \in \{0, 1, 2, 3\}$. maxpool is the max-pooling operator. Decoder aims to gradually restore the image until the resolution of the output image is the same as HR HSI. There is one deconvolution layer with a stride size 2 and two convolution layers in each stage of the decoder. The formula is given by

$$Dd_i = \begin{cases} \text{decov}\left(\sigma\left(W_i^2 * \sigma\left(W_i^1 * Dd_{i-1}\right)\right)\right), & i \in \{0, 1, 2\} \\ W_i^1 * Dd_{i-1}, & i = 3 \end{cases} \tag{8}$$

where decov is deconvolution to up-sample the features. Dd_i is the output and W_i^1 and W_i^2 are the first and second convolutions in ith stage of the decoder, respectively. $i \in \{0, 1, 2, 3\}$. Dd_{-1} is Ee_3. When $i = 3$, Dd_3 is the output of the detail extraction network.

C. Spatio-Spectral Fusion Network

The SSFN adopts a similar structure to the detail extract network, constituting the dual network, to better integrate the information from the detail extract network. The architecture of SSFN is displayed in Fig. 1, that is, blue part. The fusion process contains three steps, that is, direct fusion, slow fusion, and late fusion, effectively improving the generality of the network to different scenes.

1) Direct Fusion: MSI and HSI are concatenated along the spectral channel and then input into MSSAM. After MSI and HSI are input into the attention module, and two convolution layers with 64 convolution kernels are employed to further extract and fuse image features. Through this step, many fundamental spatial information from MSI can be directly extracted, which can help network repair details to a great extent. The formula is given by

$$Ue_0 = \sigma\left(W_0^2 * \sigma\left(W_0^1 * F\right)\right) \tag{9}$$

where F is the output of MSSAM, Ue_0 is the output of direct fusion, W_0^1 and W_0^2 are two convolutions, and $*$ is the convolution operator. The convolutions of SSFN are of size 3×3 which are denoted by the blue bar.

2) Slow Fusion: The scenes of remote-sensing images are very complex and the semantic information is difficult to maintain. Therefore, for enriching spatial details, features of each stage in the detail extraction network are injected into HSI as shown in Fig. 1. It is significant for the network to pay attention to different types of objects. Hence, during slow fusion, down- and up-sampling operators in different network depths produce multiscale and hierarchical features. Meanwhile, different spatial information from the detail extraction network are constantly compensated for the features of SSFN. The max-pooling with a stride size 2 is used to down-sample the features in each stage of the encoder. Max-pooling is denoted by a green bar. In the encoder of SSFN, three down-sampling operators with convolutions are performed to catch context information. The formula is denoted by

$$Ue_n = \sigma\left(W_n^2 * \sigma\left(W_n^1 * \text{maxpool}(\text{cat}(Ue_{n-1}, E_{n-1}))\right)\right) \tag{10}$$

where Ue_n is the output from each stage of the encoder, W_n^1 and W_n^2 are the first and second convolutions of nth stage, and E_{n-1} is the output of $n-1$th stage in the detail extraction network. $n \in \{1, 2, 3\}$.

The decoder gradually restores the size of features and more details. To catch context information, there is a skip connection to receive features from the encoder in each stage in the decoder of SSFN. Except that, each stage in the decoder of SSFN also receives the features from the previous stage in the decoder of both networks. These features of each stage are fused by two convolutions and then up-sampled by a deconvolution size of 2×2 with a stride size 2. The deconvolution is denoted by a yellow bar. Number of each deconvolution output channel is only half as that of the input channels. After the last deconvolution layer, one convolution layer concatenates and fuses shallow features, up-sampled features from the decoder and detail extraction network for the output of SSFN. The channel number of the output equals to that of HSI. The formula is given by

$$Ud_i = \begin{cases} \text{decov}\left(\sigma\left(W_i^2 * \sigma\left(W_i^1 * \text{cat}(Ue_3, Ee_3)\right)\right)\right), & i = 0 \\ \text{decov}\left(\sigma\left(W_i^2 * \sigma\left(W_i^1 * \text{cat}(Ue_{3-i}, Ee_{i-1}, Ud_{i-1})\right)\right)\right), & i \in \{1, 2, 3\} \end{cases} \tag{11}$$

where Ud_1 represents the output features of the first up-sampling stage. decov is the deconvolution which ensures that

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the sampling ratio is 2 and the number of feature channels is reduced by half. W_i^1, W_i^2, and Ud_i are the two convolution layers with kernel size 3×3 and the output of ith up-sampling stage, respectively. $i \in \{0, 1, 2, 3\}$, when $i > 0$, there are three concatenated feature blocks from both networks. Ud_3 is the output of SSFN.

3) Late Fusion: Up-sampled HSI and outputs of both networks are added to restore high-frequency information by pixel-level addition instead of concatenation. This step can guide the detail extraction network to extract the high-frequency information at target resolution and accelerate the training of whole network by a skip connection. This strategy also plays a role in reconstruction, which is proved in the ablation study. The late fusion stage is given by

$$H_f = H_{\text{up}} + U_r + E_r \tag{12}$$

where H_f is the reconstruction image.

D. Loss Function

In the proposed method, a global residual connection urges the network to restore high-frequency details. $L1$ loss is more suitable for dealing with high-frequency problems. Compared with $L2$ loss, $L1$ loss can tolerate abnormal values and achieve faster converge speed. Although $L2$ loss tends to get higher peak signal to noise ratio (PSNR), it may result in the worse visual performance and over-smooth in reconstruction images. Therefore, it is better to choose $L1$ loss for guiding the whole network to fit complex relationship between the degradation image and the truth image. The formula is written as

$$L_1 = \frac{1}{H \times W \times B} \sum_{i=1,j=1,k=1}^{H,W,B} |H_f(i,j,k) - H_R(i,j,k)| \tag{13}$$

where $i, j,$ and k represent location of each pixel. Multispectral and hyperspectral fusion is a 3-D problem involving spatial and spectral reconstruction, which not only need to repair the spatial details, but also to ensure the accuracy of the spectrum as much as possible. There is some high-frequency information in the spectral gradient, hence a spectral gradient loss is introduced to help network keep a watchful eye on spectral reconstruction. The formula is written as

$$L_S = \frac{1}{H \times W \times (B-1)} \sum_{i=1,j=1,k=1}^{H,W,B-1} \big|\big(H_f(i,j,k+1) - H_f(i,j,k)\big) - \big(H_R(i,j,k+1) - H_R(i,j,k)\big)\big| \tag{14}$$

where L_S is the spectral gradient loss. $L1$ loss is exploited to restore spectrum as mentioned before.

The whole loss function is written as

$$L = \alpha L_1 + \beta L_S \tag{15}$$

where α and β are the weight factors of loss functions.

III. Experiments

The visual and quantitative results of the experiments are displayed in this section. Three simulation experiments and one real experiment validate the effectiveness of the proposed method, compared with the traditional and other deep-learning-based methods. Meanwhile, the analysis of the experimental results is also given. Beyond that, the ablation study is conducted to prove that each component of the network plays a role in the improvement of performance.

A. Datasets in Simulation Experiments

Three common and open datasets are adopted to conduct the comparison experiments, including 1) 2018 IEEE Contest Dataset, 2) Washington dc Mall Dataset, and 3) University of Pavia Dataset. The experiment results verify the satisfying performance and generality of the proposed method. For the datasets below, due to the unavailability of the ground-truth image in the realistic world, we regard the original HSI as the truth IEEE Contest Dataset: The dataset is provided by The Standardized Remote Sensing Data Website of the IEEE Geoscience and Remote Sensing Society (GRSS) for 2018 IEEE GRSS Data Fusion Contest. For HSI, it includes 50 bands whose size is 512×512. The related MSI is the red-green-blue (RGB) image whose size is down-sampled to 512×512.

1) Washington dc Mall Dataset: Washington dc Mall Dataset (third dataset): This dataset is collected by the HYDICE sensor; the size is 1280×307; the number of bands is 191, which covers 400–2500 nm; the spatial resolution is 2.8 m.
2) University of Pavia Dataset: The dataset is collected over the area of Pavia, Northern Italy, by the ROSIS sensor. After 12 noisy bands are removed in the experiment, 103 spectral bands are adopted. The spatial resolution is 1.3 m, and the size of the image is 610×340.

B. Evaluation Metrics

For quantitative assessment, four commonly adopted metrics describe the quality of the images which help with the comparison of the methods in this article. The first metric is PSNR which evaluates the spatial quality of the image with band as unit. The higher the PSNR, the better the performance is. Another spatial metric is structural similarity (SSIM) [40] which evaluates the image quality based on the principle that human visual system (HVS) is sensitive to local structure changes. It is a perceptually motivated metric. The higher the SSIM is, the better the performance is. The best is 1. The third metric is to evaluate the spectral preservation of the image, spectral angle mapper (SAM) [41]. The smaller the index of SAM, the better the performance is. Final metric is erreur relative globale adimensionnelle de synthèse (ERGAS) [42] which is utilized to assess the global statistical quality of the fused image. The smaller ERGAS, the better the performance is.

C. Experimental Settings

In this section, the experimental settings are introduced. Since the true data pairs for training are difficult to acquire, Wald protocol [40] is applied to conduct simulation experiments. The original HSI is regarded as the truth image, hence down-sampled HSI is the LR HSI. MSI is sampled to the same resolution as LR HSI. In the simulation experiments, the means of the down-sampling operator is bicubic and the spatial down-sampling factor is 4.

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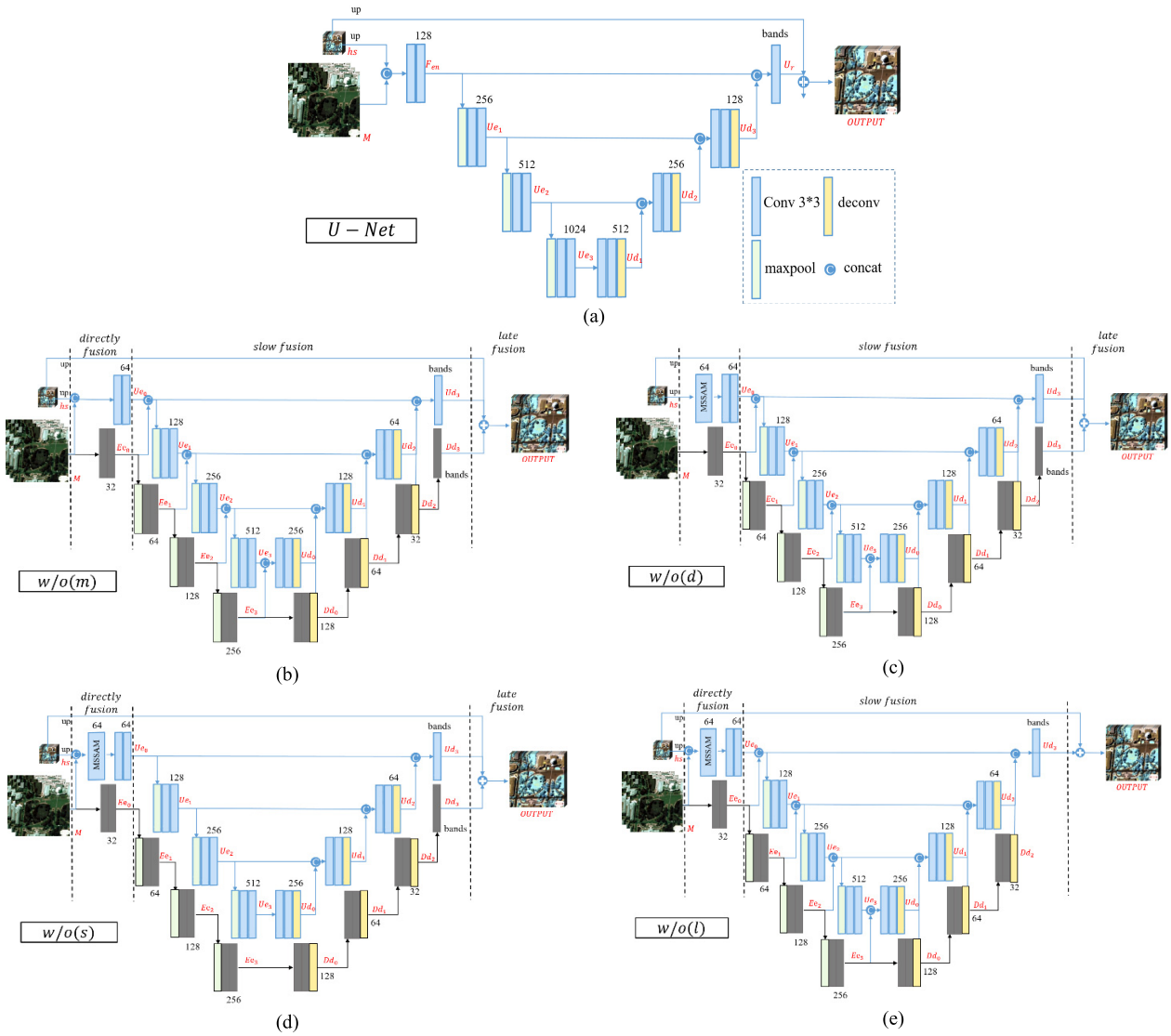


Fig. 4. Visual comparison on the real dataset. (a) U-Net. (b) w/o(M). (c) w/o(d). (d) w/o(s). (e) w/o(l).

1) For 2018 IEEE Contest Dataset, the whole remote-sensing images including MSI and HSI are divided into 14 parts. The left 12 pairs of images are used for training and the remaining area is for testing. The 5292 training patches, cropped from the training area, are of size 32×32, for example, the sizes of LR HSI, MSI, and HR HSI are $8 \times 8 \times 50$, $32 \times 32 \times 3$, and $32 \times 32 \times 50$, respectively. The image patches are randomly selected from the training data in each iteration of training. The sizes of MSI and HSI are $512 \times 512 \times 3$ and $512 \times 512 \times 50$, respectively.
2) For the Washington dc Mall dataset, the training patches are cropped from the upper part of the image $1152 \times 307 \times 191$. The number of image patches whose sizes are 16×16 is 8437. Due to unavailability of true MSI, the MSI is generated by a uniform spectral response function corresponding to a specific set of Landsat TM bands. The sizes of LR HSI, MSI, and HR HSI are $4 \times 4 \times 191$, $16 \times 16 \times 6$, and $16 \times 16 \times 191$, respectively. The remaining area is for test. The sizes of MSI and HSI for test are $128 \times 128 \times 191$.
3) For the University of Pavia Dataset, the upper part $482 \times 307 \times 103$ is used for training while the remaining area is for testing. The settings are the same as in Washington dc Mall dataset.

The network is trained and tested under the Pytorch framework with GPU 2080Ti. As for the weight factors in loss function, both α and β are set as 1, according to our experience, to make the two values on the same level of magnitude. During the training process, the Adam optimizer is used to train the network. The initial learning rate is set as 0.0002, which will decrease to 0.8 times as the previous value for each 10 epochs. For a fair comparison, all networks are trained for 200 epochs. The whole training time on three datasets is about 8–9 h.

D. Performance Comparison

This section mainly explores the roles of the components of D-UNet. D-UNet with different components, that is, different variants, are trained on the same training data, Washington dc Mall dataset. Fig. 4(a) is the U-Net, Fig. 4(b) is w/o(M), Fig. 4(c) is w/o(d), Fig. 4(d) is w/o(s), and Fig. 4(e) is w/o(l). For validating the effectiveness of the proposed attention

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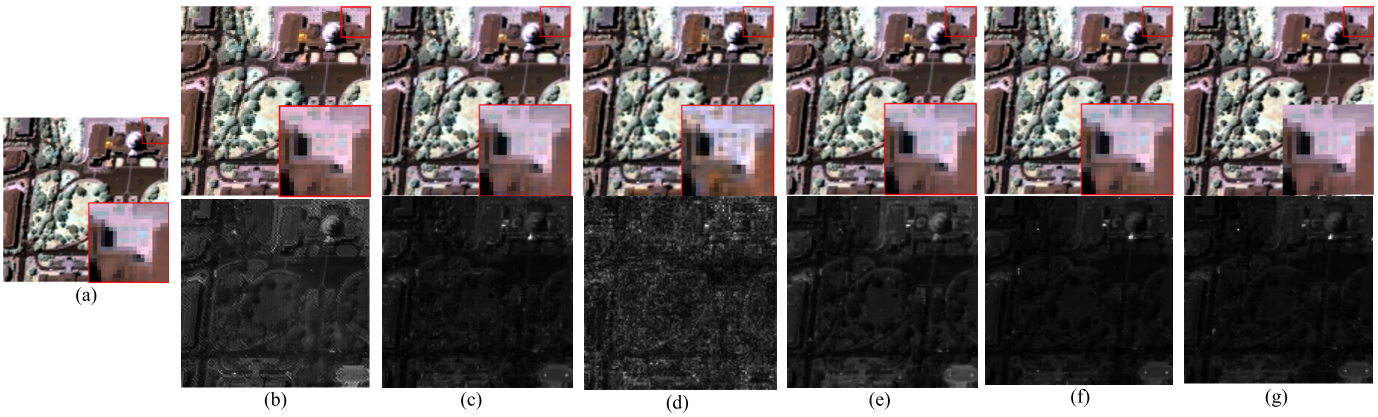


Fig. 5. Residual graphs and fusion results on the Washington dc Mall dataset. The fusion results are displayed with bands 100-80-60. (a) Truth image. (b) U-Net. (c) w/o(M). (d) w/o(d). (e) w/o(s). (f) w/o(l). (g) D-UNet. (b)–(g) Color fusion images on the top and the residual graphs on the bottom of the competing networks.

module, the first variant removes the MSSAM, compared with the original network, denoted by w/o(M). In order to verify the effectiveness of three fusion steps, three variants are specially designed. The second variant removes the direct fusion, denoted by w/o(d). The third variant removes the slow fusion, denoted by w/o(s), while the fourth variant removes the late fusion, that is, w/o(l). Except that, an original U-Net is set as baseline to compare with the networks of different architectures and demonstrate the effectiveness of each component, notably because the parameter number of the original U-Net is relatively small compared to the proposed method. In order to make a fairer comparison, the number of convolution kernels at each layer of U-Net is increased. The architectures of five variants are shown in Fig. 4.

To analyze the performances of various networks, qualitative and quantitative evaluation results on the Washington dc Mall dataset described in Section III-A are given together. The reconstructed images of all variants which are displayed with bands 100-80-60 corresponding RGB band, respectively, are shown in Fig. 5. The lower-right region represents the magnified show of the image inside the red rectangle. Fig. 5(a) is the truth image. Fig. 5(b)–(g) shows the MSI and HSI fusion results of U-Net, w/o(M), w/o(d), w/o(s), w/o(l), and D-UNet, respectively. In order to better evaluate the fusion performance of each network, the residual graphs of them and the real image is also shown at the bottom of the fusion result diagrams. For the residual graph, the formula becomes

$$\text{Res} = \sum_{b=1}^{B} |H_{fb} - H_{Rb}|/B \tag{16}$$

where B is the band number of HSI. It is helpful to observe the spatial details restored by the network through residual graphs. It is noted that the residual graphs displayed in Fig. 5 are normalized. The better the network reconstructs HR HSI, the less image content will be left in the residual graph. Hence, the value of pixels in better reconstruction result is closer to zero, which leads the gray image to darker. It is seen that the residual graph of D-UNet contains less content than other variants. For U-Net, its fusion result is over-smooth. Obviously, even if the parameter number of U-Net is doubled, its performance is not as good as that of D-UNet. The residual graph also indicates that it is helpful to restore high-frequency information with the effective spatial information compensation. When MSSAM is removed, the spectrum of the fusion image is distorted and the color changes of materials in the magnified image are also unnatural. When the first fusion step, that is, direct fusion, is removed, the performance is very poor and the fusion image is too vague. There is an obvious spatial and spectral distortion in the result of w/o(d). Similarly, the absence of slow fusion also leads to a poor reconstruction result. Especially, there are some abrupt bright spots in the zoom area of w/o(s). And the residual figure of w/o(s) also shows that there is still a lot of missing information. For the results of w/o(l), the areas of the image where the color changes will have some distortion. Compared the residual graph of D-UNet with that of w/o(l), it is demonstrated that late fusion is also helpful for restoration. The quantitative evaluation is presented in Table I. As seen from the table, D-UNet is the best architecture. Because MSSAM can help the network to comprehensively consider different types of objects from the perspective of multi-scale and improves accuracy of reconstruction. When direct fusion is removed, the injected spatial features from the detail extraction network lack basics but important details of MSI which are lost in the max-pooling and convolution layers. In consequence, the network cannot acquire enough and efficient spatial information which leads to the bad performance. During slow fusion, most multiscale and hierarchical spatial features of the detail extraction network are provided. This step is designed to extract rich semantic information of MSI. Because various spatial features are embedded in the SSFN during each sampling operator, the network can further repair different spatial details stage by stage. Therefore, the performance of w/o(s) is also not as good as D-UNet. For late fusion, it is responsible to inject the output of the detail extraction network which can restore high-frequency information with the size of the target resolution. It can be seen from the residual graphs of w/o(l) and D-UNet that this step plays a certain role in the reconstruction of the mutated region.

E. Simulation Experiments

To prove that D-UNet has a good ability to fuse MSI and HSI, three simulation experiments conducted on the datasets.

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TABLE I
PERFORMANCE ON WASHINGTON DC MALL DATASET

	PSNR	SSIM	SAM	ERGAS
U-Net	35.5951	0.9980	1.4884	1.3351
w/o(M)	35.3145	0.9952	1.4332	1.6676
w/o(d)	32.9415	0.9777	2.5108	3.6059
w/o(s)	34.5429	0.9949	1.3227	2.3068
w/o(l)	35.6164	0.9983	1.4126	1.5615
D-UNet	**36.8401**	**0.9986**	**1.2331**	**0.9862**

TABLE II
PERFORMANCE ON IEEE GRSS DATASET

	PSNR	SSIM	SAM	ERGAS
CNMF	28.0574	0.95542	9.6328	3.7679
3D-CNN	32.8391	0.9607	2.7153	1.4465
PFCNN	33.2356	0.9626	3.2112	1.3967
TF-Net	33.2534	0.9655	2.6812	1.2976
MSDCNN	33.5949	0.9739	2.6737	1.2877
PROPOSED	**34.3145**	**0.9790**	**2.5439**	**1.2441**

TABLE III
PERFORMANCE ON WASHINGTON DC MALL DATASET

	PSNR	SSIM	SAM	ERGAS
CNMF	29.0766	0.9717	8.7784	6.5786
3D-CNN	32.5462	0.9826	2.9737	1.8036
PFCNN	33.1922	0.9837	3.6874	1.9967
TF-Net	33.5949	0.9839	3.1521	1.2976
MSDCNN	34.3814	0.9897	2.6990	1.2422
PROPOSED	**36.8401**	**0.9986**	**1.2331**	**0.9862**

TABLE IV
PERFORMANCE ON UNIVERSITY OF PAVIA DATASET

	PSNR	SSIM	SAM	ERGAS
CNMF	28.5716	0.9024	12.4754	6.8774
3D-CNN	29.0838	0.9128	9.5485	5.6874
PFCNN	29.3434	0.9283	5.9992	7.5846
TF-Net	28.4748	0.9131	7.4561	3.2366
MSDCNN	30.3645	0.9322	5.1521	3.1323
PROPOSED	**31.7114**	**0.9449**	**4.3009**	**2.5426**

D-UNet will compete with six state-of-the-art methods including traditional and deep learning algorithms. These methods are coupled non-negative matrix factorization (CNMF) [29], 3-D CNN (3D-CNN) [22], pyramid fully CNN (PFCNN) [19], two-stream fusion network (TF-Net) [30], and SSR-NET [21]. CNMF belongs to matrix factorization-based method which uses endmember matrix of HSI and abundance map of MSI to reconstruct HR HSI. Other four methods are deep learning, but each method owns different contributions. 3D-CNN utilizes three 3-D convolution layers to fuse MSI and principal components of HSI obtained by principal component analysis (PCA). PFCNN also applies a U-Net where the MSI down-sampled by bicubic at different multiples are added to the decoder part. TF-Net is based on the architecture of U-Net with a two-stream strategy in the front of the network. SSR-NET is a three-layer network that uses 2-D convolution for fusion where the second and third layers are equipped with residual connections. One interesting point of SSR-NET is that the channel number of all convolutions is equal to that of HSI. The fusion images are displayed in Figs 6–8 and quantitative results are listed in Tables II–IV.

1) Results on 2018 IEEE Contest Dataset: Fig. 6 shows the fusion images with bands 45-30-15 which are randomly selected from the test images. Traditional methods pay more attention to spatial restoration. The result of CNMF is visually sharper than that of the deep learning-based methods, but there are noticeable spatial artifacts and aliasing in the zoomed image. There is an obvious spectral distortion, compared with the deep learning methods. 3D-CNN obtains a better result which is more continuous in space with reduced spectral distortion. However, it suffers from blurring effects as the edges of objects in the zoomed image of Fig. 6(d). Also, the residual graph indicates that many details in this image have not been repaired. PFCNN presents a distinct color distortion and blurring effects, as it can be observed that the missing information is noticeable according to its residual graph. TF-Net and SSR-NET achieve visually better fusion results with reduced spectral distortion and blurring effects. As seen in Fig. 6, TF-Net has clearer spatial details and better spectral preservation. There is also less missing information in the residual graph. From Fig.6, D-UNet can achieve the best performance whose spatial structure is clearest and the spectrum is closest to the truth image. The residual graph is darker than others as well. Table II lists the values of four indicators of the competing methods, with average of all test images. In Table II, SSR-NET achieves the second best performance in all indicators. Both TF-Net and PFCNN acquire higher PSNR and SSIM with lower ERGAS, but higher SAM than 3D-CNN. Maybe, 3D-CNN employs 3-D convolution to fuse images which can take spectral consistence into consideration.

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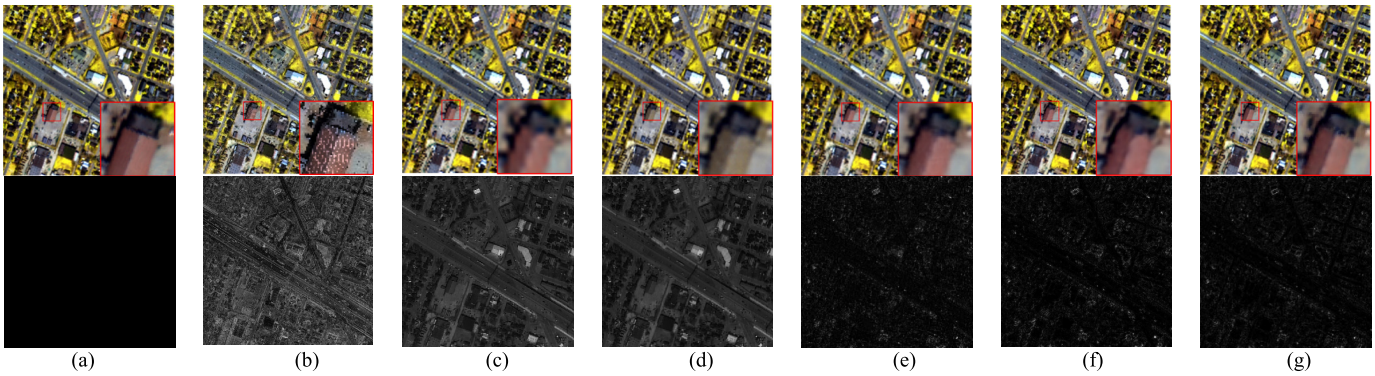


Fig. 6. Residual graphs and fusion results on the 2018 IEEE Contest dataset. The fusion results are displayed with bands 45-30-15. (a) Truth image. (b) CNMF. (c) 3D-CNN. (d) PFCNN. (e) TF-Net. (f) SSR-NET. (g) D-UNet. (b)–(g) Color fusion images on the top and the residual graphs on the bottom of the competing networks.

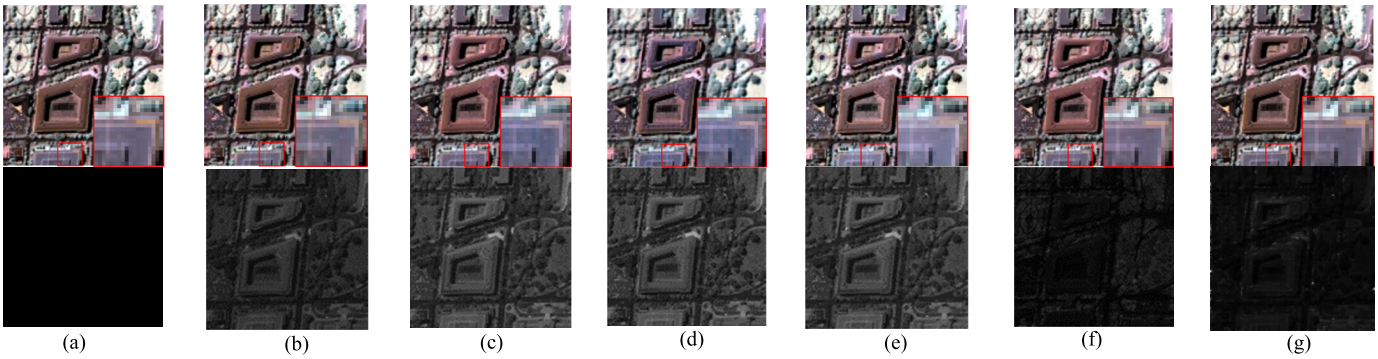


Fig. 7. Residual graphs and fusion results on the Washington dc Mall data set. The fusion results are displayed with bands 100-80-60. (a) Truth image. (b) CNMF. (c) 3D-CNN. (d) PFCNN. (e) TF-Net. (f) SSR-NET. (g) D-UNet. (b)–(g) Color fusion images on the top and the residual graphs on the bottom of the competing networks.

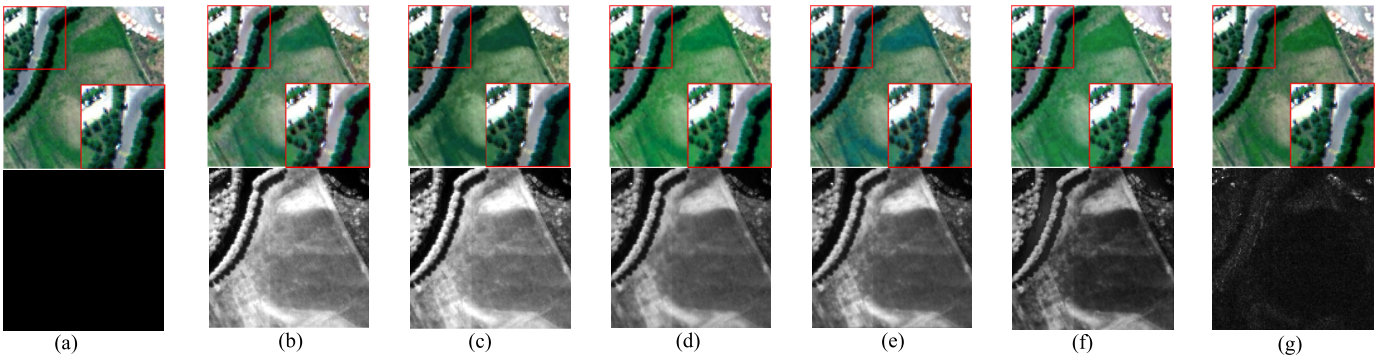


Fig. 8. Residual graphs and fusion results on the Washington dc Mall dataset. The fusion results are displayed with bands 45-30-15. (a) Truth image. (b) CNMF. (c) 3D-CNN. (d) PFCNN. (e) TF-Net. (f) SSR-NETm. (g) D-UNet. (b)–(g) Color fusion images on the top and the residual graphs on the bottom of the competing networks.

Nonetheless, PCA destroys the image information, leading to the missing information. On the contrary, TF-Net and PFCNN pay more attention to spatial details compensation, both of which apply architecture of U-Net to integrate different spatial features.

2) Results on Washington dc Mall Dataset: The second experiment is conducted on the Washington dc Mall dataset. The fusion results and residual graphs are presented in Fig. 7. The color images are shown with bands 100-80-60 corresponding to the RGB band. For comparing the details of these images, the upper-right region represents the magnified show of the image inside the red rectangle. According to the fusion images, it is observed that the traditional methods do not have sharper spatial information than the deep learning-based methods. Specifically, no significant increase in spatial details exists in the vegetation and buildings areas of fusion results. In addition, the objects in the images of both PFCNN and TF-Net have an obvious spectral distortion. Both SSR-NET and D-UNet achieve the best visual effects. But, there is less missing information left in the image of D-UNet according to

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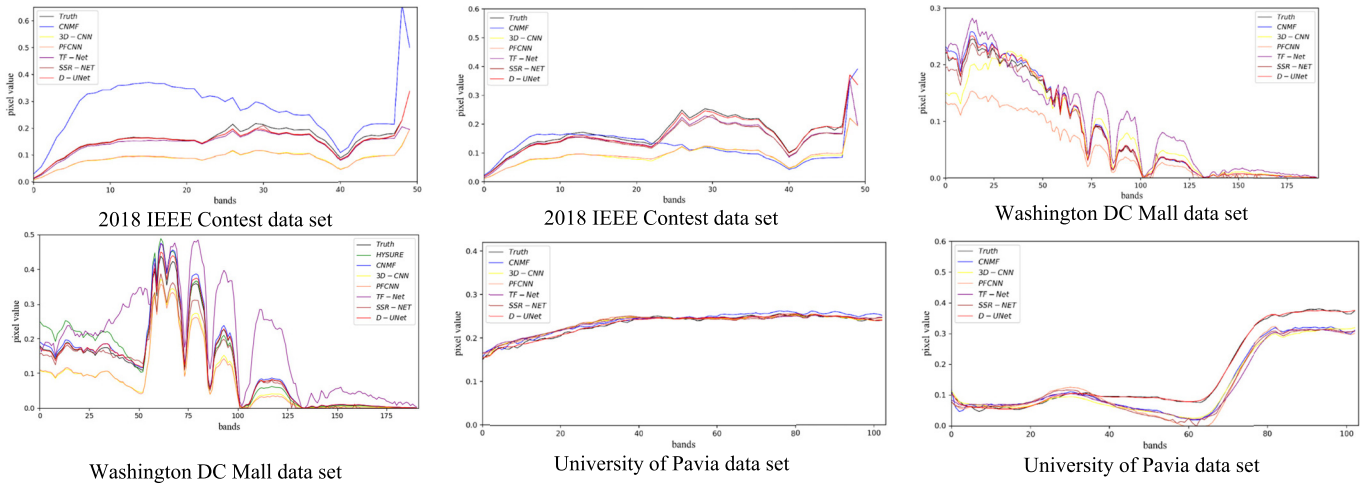


Fig. 9. Spectral curve of results from the competing methods at some random points on the datasets. The column on the left represents urban land, and the column on the right represents vegetation.

the residual graphs. Table III shows that, among the methods above, the deep learning-based methods outperform two traditional methods in all indicators. Moreover, 3D-CNN performs better in spectral fidelity, while performs worse in spatial enhancement than PFCNN. For TF-Net, it uses a two-stream strategy to extract spatial features pertinently, followed by a U-Net with residual connections which can help network acquire a better fusion result. SSR-NET achieves the second best performance in the four indicators. Notably, the spatio-spectral edge loss helps SSR own a superior performance to PFCNN even if with a shallow network. Because D-UNet takes full advantages of multiscale and hierarchical features and properly injects the spatial features under the condition of maintaining the spectrum as much as possible, it achieves the best fusion performance in spectral and spatial fields.

3) Results on University of Pavia Dataset: The last simulation experiment is conducted on the University of Pavia dataset. The fusion results and related residual graphs of the methods are displayed in Fig. 8 which are randomly selected from the test images. The color images are shown with bands 45-30-15 corresponding to RGB bands where the lower-right corner shows a magnified display of the image inside the red rectangle. It is seen from Fig. 8 that spectral distortion exists in different degrees in each fused image. Fig. 8(b)–(h) displays the residual images between the fusion results of the above methods and truth image, and Fig. 8(a) is the reference residual image. The residual image of PFCNN is similar to TF-Net, but more ambiguous especially in the mixture of vegetation and bare soil. It is observed from the residual graphs that SSR achieves better performance on vegetation area but worse performance on river area, which is different from the last experiment. Maybe SSR owns too few layers to adapt to all scenes. Compared with the upper-right corner of the truth image including white and red features, both traditional and other deep learning-based methods can hardly remain red features. Only D-UNet can reconstruct the details and obtain the most similar spectral characteristics to the reference image.

Table IV lists the quantitative evaluation indicators of the University of Pavia dataset. It is found that D-UNet achieves the best performance in terms of four quantitative indicators. Traditional methods perform poorly in the indicators while deep learning-based methods exhibit superior performance. In this dataset, PFCNN has higher PSNR and SSIM with lower SAM than TF-Net which is different from previous experiments. Perhaps, the architecture of PFCNN is more suitable to the University of Pavia dataset than TF-Net. In addition, to analyze the spectral preservation of the competing methods, spectral curves at some random points in the fusion results for all methods on the three remote-sensing datasets are shown in Fig.9. The images in the left column show the spectral curve of the urban land on three datasets, while those in the right column show the spectral curve of the vegetation on three datasets. The red line represents the spectral curve of D-UNet and the black line represents that of the truth image. It is observed that the spectral curve of D-UNet is closer to the truth image than other competing methods. Both traditional methods perform worse than deep learning-based methods for comparison on the 2018 IEEE Contest dataset. For the Washington dc Mall dataset, TF-Net has a bad spectral curve between 25 and 50 bands, which has an obvious different trend compared with the truth image. The spectral curves of different methods are similar on the urban land of the University of Pavia dataset. But there is an outlier in the 70th band of result obtained from PFCNN. In contrast, D-UNet also applies a two-branch architecture but achieves better performance in different areas of three data sets.

F. Real Data Experiment

To verify the effectiveness of D-UNet in practical application, real experiment is conducted on two open remote-sensing data sets. The images from Hyperion are regarded as HSI with 30 m spatial resolution and 115 bands. For MSI, we select the images from Sentinel-2. Only the bands whose spatial resolution is 10 m are regarded as MSI, that is, MSI has four bands. Due to the unavailability of real image pairs, we divide

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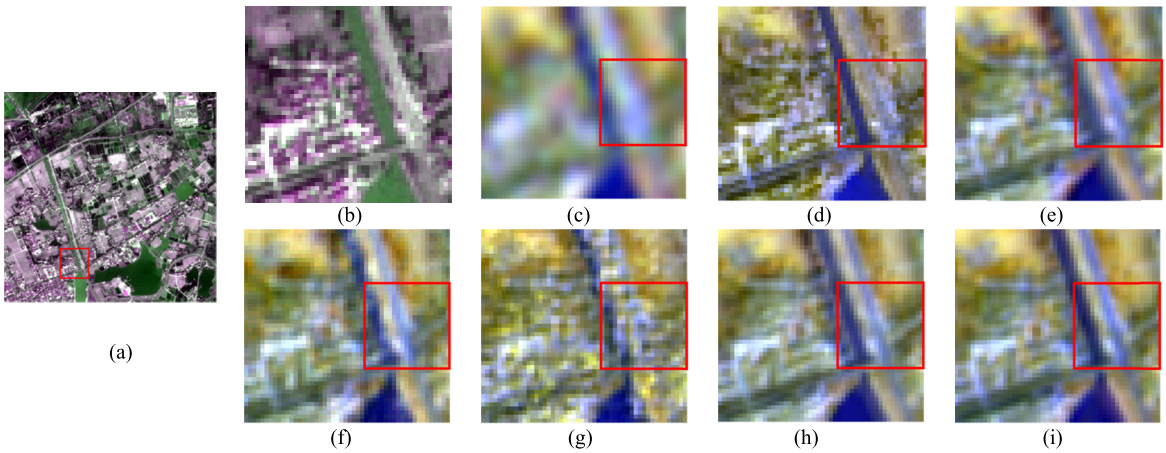


Fig. 10. Zoomed building area of the Hyperion and Sentinel-2 dataset. The fusion results are displayed with bands 80-61-20. (a) MSI. (b) Zoom. (c) Bicubic. (d) CNMF. (e) 3DCNN. (f) PFCNN. (g) TF-Net. (h) SSR-NET. (i) D-UNet. (b) Zoom image of MSI. (c)–(i) Color fusion images of the competing networks.

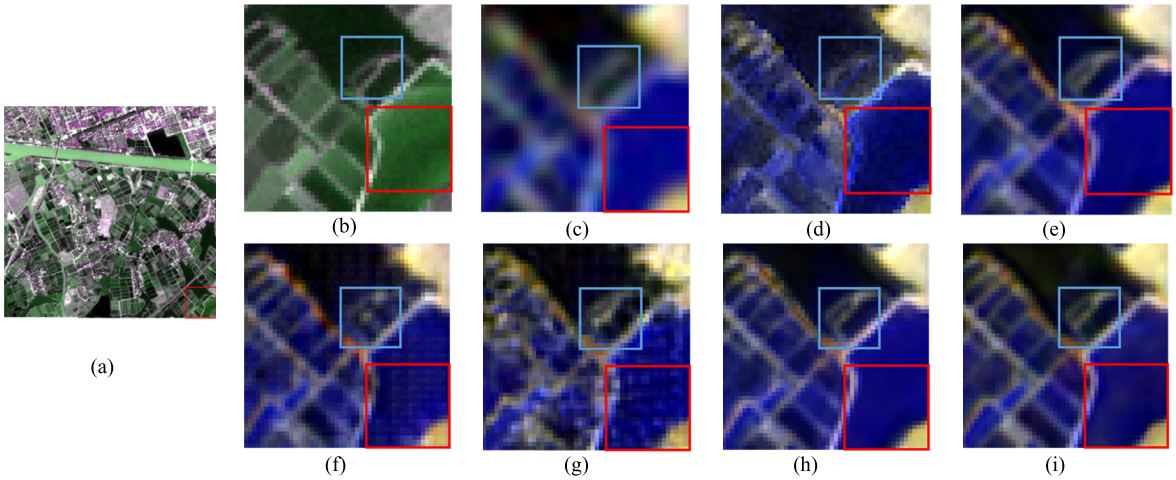


Fig. 11. Zoomed farm and water area of the Hyperion and Sentinel-2 dataset. The fusion results are displayed with bands 80-61-20. (a) MSI. (b) Zoom. (c) Bicubic. (d) CNMF. (e) 3DCNN. (f) PFCNN. (g) TF-Net. (h) SSR-NET. (i) D-UNet. (b) Zoom image of MSI. (c)–(i) Color fusion images of the competing networks.

the images into two non-overlapped parts, one of which is for training. Fourteen cloud-free image pairs with different sizes are obtained in total. In the training process, the original HSIs are regarded as truth images and MSI are down-sampled three times by bicubic to keep the same resolution as HSI. We crop image patches size of $16 \times 16 \times 115$ from HSIs as truth images and patches size of $16 \times 16 \times 4$ as MSIs. Then, the LR HSI is estimated by down-sampling original HSIs four times by bicubic. The training settings are the same as the previous ones. Adam is employed to optimize the network with 0.0002 learning rate and network is trained for 200 epochs. In the testing process, the remaining areas are used as test samples. The HSIs of the test images are cropped to 128×128 and related MSIs are cropped to 384×384. In order to further observe the fusion results, some groups of zoomed reconstruction images including different scenes are displayed in Figs. 10 and 11. Fig. 10 displays the zoomed building and river area which is marked by a red rectangle in Fig. 10(a) with bands 80-61-20. Fig. 10(b)–(i) shows the zoomed image of different methods. Compared with (b), there are various degrees of color distortions in the image of CNMF as shown in Fig. 10(c) and (d). It is seen that PFCNN and TF-Net reconstruct the images with worse spatial details than other deep learning-based methods. The image of 3D-CNN is closer to LR HSI in color than SSR-NET but spatial details are decreased. Compared with other methods, D-UNet achieves the best performance. The river way of D-UNet image is clearest among deep learning methods and D-UNet restores the paths near the river which is lost in the CNMF image as shown in the red rectangle. Fig. 11 shows the zoomed farm and water area of the Hyperion and Sentinel-2 dataset. The severe distortion appears in the image of CNMF with spatial aliasing. Also, the spatial information is destroyed in the images of PFCNN and TF-Net. Vegetation areas are green and black in the LR HSI that is shown in the blue rectangle, but those in the images of 3D-CNN and SSR-NET are only black as shown in the top part. D-UNet reconstructs most spatial details, like ponds and paths, and maintains closer color to LR HSI.

IV. Conclusion

This article designed a dual multiscale hierarchical network with a novel spatio-spectral attention module for the remote-sensing image fusion. The network uses two similar architectures based on U-Net, one of which is called SSFN and another

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is called the detail extraction network. The detail extraction network consistently compensates rich spatial features with different scales and depths for SSFN during three fusion steps. In addition, the detail extraction network consistently compensates rich spatial features with different scales and depths for SSFN during three fusion steps. In addition, a novel MSSAM is employed to further make use of multiscale information as much as possible. Meanwhile, MSSAM in the front of the network can preliminary fuse the MSI and HSI and help network pay more attention to important features. Benefiting from the dual architecture and MSSAM, D-UNet can enhance spatial details of the reconstructed images while preserving the spectrum as much as possible. The simulation and real experiments conducted on four remote-sensing data sets demonstrate that D-UNet is superior to state-of-the-art methods.

In the future, we attend to make network more interpretable and further improve the performance. Variational model-based methods can describe the degenerate relationships between images and own a derivable iterative process with mathematical significance. Therefore, they are considered to be combined with deep learning, which can help construct network. In addition, there are a number of images from various sources. It is significant to design a general network which is adapt to different data sets.

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